

SYNTHETIC CASE — FICTIONAL PATIENT. This document is entirely fabricated for a software engineering assessment. It contains no real patient data and must not be used for any clinical purpose.

Case Referral for Multidisciplinary Council Review

St. Mary's Fictional General Hospital | Case Ref: LLMC-2026-0417 | Referral Date: 14 July 2026

1. Patient Profile

Field	Detail
Name	Mr. K. R. Varma
Age / Sex	76 years / Male
Height / Weight / BMI	168 cm / 61 kg / 21.6 (unintentional weight loss of ~5 kg over 8 months)
Living situation	Lives with spouse; daughter (primary contact) lives nearby; independent for basic activities, needs assistance for shopping and stairs
Smoking / Alcohol	Ex-smoker, 30 pack-years, quit 12 years ago; no alcohol
Occupation	Retired school teacher

2. Presenting Condition & History

Mr. Varma presented to the cardiology outpatient clinic with a 6-month history of progressive exertional breathlessness (now NYHA Class III — symptomatic on walking ~100 meters or one flight of stairs), two episodes of exertional chest tightness, and one witnessed syncopal episode three weeks ago while climbing stairs, resulting in a minor fall without fracture.

Evaluation has confirmed severe symptomatic aortic stenosis with concomitant single-vessel coronary artery disease. The syncopal episode and worsening functional class indicate the disease has entered a high-risk symptomatic phase. Without intervention, symptomatic severe aortic stenosis of this profile carries a substantially elevated 1–2 year mortality risk.

3. Comorbidities

- Chronic kidney disease, stage 3b — baseline creatinine 1.9 mg/dL, eGFR 34 mL/min/1.73m², stable over the last 12 months; followed by nephrology; no prior dialysis.
- Type 2 diabetes mellitus — 18 years, on oral agents; HbA1c 7.8%; background diabetic retinopathy; likely contributor to nephropathy.
- Chronic obstructive pulmonary disease (moderate) — FEV1 55% predicted; one exacerbation requiring oral steroids in the past year; on inhaled bronchodilator therapy; no home oxygen.
- Prior transient ischaemic attack — 3 years ago, full recovery; carotid Doppler shows 50% right internal carotid stenosis (asymptomatic since).
- Hypertension — controlled on two agents.

6. Mild chronic anaemia — Hb 10.8 g/dL, attributed to CKD; iron studies borderline.
7. Borderline frailty — see Section 6.

4. Current Medications & Allergies

Medications

- Metformin 500 mg twice daily (dose-reduced for renal function)
- Gliclazide 40 mg once daily
- Amlodipine 5 mg once daily
- Telmisartan 40 mg once daily
- Aspirin 75 mg once daily
- Atorvastatin 40 mg at night
- Tiotropium inhaler once daily; salbutamol inhaler as needed
- Oral iron supplementation

Allergies

- Documented allergy to iodinated contrast media — previous moderate reaction (urticaria and bronchospasm) during a CT scan 5 years ago, managed without intensive care. Premedication protocol required for any contrast exposure.
- No known drug allergies otherwise.

5. Key Investigations

Laboratory results (10 July 2026)

Test	Result	Reference	Note
Creatinine	1.9 mg/dL	0.7–1.3	Stable x 12 months
eGFR	34 mL/min/1.73m ²	>90	CKD stage 3b
Urea	58 mg/dL	15–40	Elevated
Potassium	4.9 mmol/L	3.5–5.1	Upper normal
Haemoglobin	10.8 g/dL	13–17	Mild anaemia
HbA1c	7.8%	<6.5	Suboptimal control
NT-proBNP	2,450 pg/mL	<300	Significantly elevated
Albumin	3.4 g/dL	3.5–5.0	Low-normal
INR / Platelets	1.0 / 190,000	—	No coagulopathy

Transthoracic echocardiogram (08 July 2026)

- Severe calcific aortic stenosis: valve area 0.7 cm², mean gradient 46 mmHg, peak velocity 4.4 m/s.
- Left ventricular ejection fraction 45% (mild–moderate impairment); moderate LV hypertrophy.
- Mild mitral regurgitation; mild pulmonary hypertension (estimated PASP 42 mmHg).

Coronary angiography (11 July 2026, minimal-contrast protocol with premedication)

- 70% stenosis of the proximal left anterior descending (LAD) artery; other vessels with minor irregularities only.
- Procedure tolerated; creatinine 48 hours post-procedure 2.0 mg/dL, returning to baseline at day 5.

Pulmonary function tests

- FEV1 55% predicted, FEV1/FVC 0.62 — moderate obstructive pattern; adequate but reduced respiratory reserve for major surgery.

Risk scores (calculated by the cardiology team)

- STS predicted risk of mortality (SAVR + CABG): approximately 6.8% — high-intermediate surgical risk.
- EuroSCORE II: approximately 7.5%.
- Predicted risk of post-operative acute kidney injury judged “high” given CKD 3b, diabetes, anaemia, and anticipated cardiopulmonary bypass time.

6. Functional & Frailty Assessment

- Clinical Frailty Scale: 4–5 (vulnerable to mildly frail).
- Gait speed 0.7 m/s (slow); hand-grip strength below age norm; unintentional weight loss as noted.
- Cognition intact (MoCA 26/30). Mood: mildly anxious about surgery; no depression.
- Patient’s stated preference: “I want to be able to walk to the temple again and see my granddaughter’s wedding next year. I will accept surgery if the doctors believe it is the right thing, but I am afraid of ending up on dialysis.”
- Daughter (primary contact) has expressed concern about surgical risk and asked whether “the valve can be fixed without open surgery.”

7. Proposed Procedure Under Consideration

Surgical aortic valve replacement (SAVR) with concomitant single-vessel coronary artery bypass grafting (CABG to the LAD), under cardiopulmonary bypass, proposed within the next 4–6 weeks.

Alternative under discussion: transcatheter aortic valve implantation (TAVI) with staged or medical management of the LAD lesion. Key tensions the council should weigh: TAVI avoids cardiopulmonary bypass (lower peri-operative renal and pulmonary stress) but leaves the coronary lesion to be managed separately and requires contrast exposure in a contrast-allergic CKD patient; SAVR + CABG treats both lesions definitively in one operation but carries higher peri-operative risk to kidneys and lungs in this patient.

8. Consulting Cardiologist — Initial Review & Recommendation

“Mr. Varma has severe symptomatic aortic stenosis with syncope and NYHA III symptoms — the natural history without intervention is poor, and I do not consider conservative management a reasonable primary option. His coronary anatomy (proximal LAD 70%) argues for definitive combined treatment, and on anatomy alone SAVR + CABG is the most complete solution.”

My recommendation is to proceed with SAVR + CABG within 4–6 weeks, subject to council review. However, I want to flag renal function as my principal concern: he is CKD stage 3b with diabetes and anaemia, and the predicted risk of post-operative acute kidney injury — including a non-trivial risk of temporary or permanent dialysis — is high with cardiopulmonary bypass. His expressed fear of dialysis is directly relevant to consent. I request: (1) formal nephrology input on peri-operative renal protection and dialysis-risk quantification; (2) anaesthesiology assessment of fitness given COPD and frailty; (3) explicit council consideration of TAVI as the lower-renal-stress alternative, noting his contrast allergy complicates TAVI work-up and the LAD lesion would remain; and (4) a documented shared-decision conversation with the patient and his daughter before final consent.”

— Dr. Meera Krishnan, Senior Consultant Cardiologist, 14 July 2026

9. Questions Referred to the Council

8. Should the proposed SAVR + CABG proceed as recommended, be modified (e.g., TAVI-first strategy), be delayed pending further optimisation, or be declined in favour of medical management?
9. If surgery proceeds, what peri-operative renal-protection measures and contingency plans (including dialysis readiness) should be mandated?
10. Is the patient's respiratory and frailty status compatible with cardiopulmonary bypass and sternotomy recovery, and what prehabilitation is advised within the 4–6 week window?
11. How should the patient's stated goals and dialysis fears, and the family's questions about less invasive options, be weighed and documented in the consent process?
12. Are there institutional resource or scheduling constraints (ICU beds, dialysis capacity, theatre availability) that affect timing?

Note to candidates: this document is the standard input for your LLM Council. The case is deliberately balanced — there are legitimate grounded arguments for proceeding, modifying, delaying, and declining. Your system's job is not to find the “right” clinical answer; it is to surface these tensions faithfully, ground every argument in the case and your knowledge bases, record dissent, withstand your red team, and route the synthesis to a human. All names, values, and institutions are fictional.